ANDROID PROGRAMMING PRACTICAL 14

Module 1 · GridView with Custom Adapter

Aim

Implement GridView using a custom adapter.

Project name: GridViewCustomAdapterDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create an Empty Views Activity project and add a GridView to activity_main.xml.

2. Prepare a data model or arrays for item titles and optional images.

3. Create a custom item layout containing an ImageView and TextView (or title only).

4. Create a custom adapter extending BaseAdapter; implement getCount(), getItem(), getItemId(), and getView().

5. Inflate the item layout in getView(), bind the correct data, and reuse convertView where possible.

6. Set column count, spacing, and the custom adapter on GridView.

7. Run the app and verify all cells display the correct data and respond to item clicks.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

Grid items are displayed through a custom adapter and show the correct bound data.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.